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IoT-Based Patient Health Monitoring System

Engineering Clinics – Multi Disciplinary Project
ECS-201

BACHELOR OF TECHNOLOGY
(Artificial Intelligence and Data Science.)



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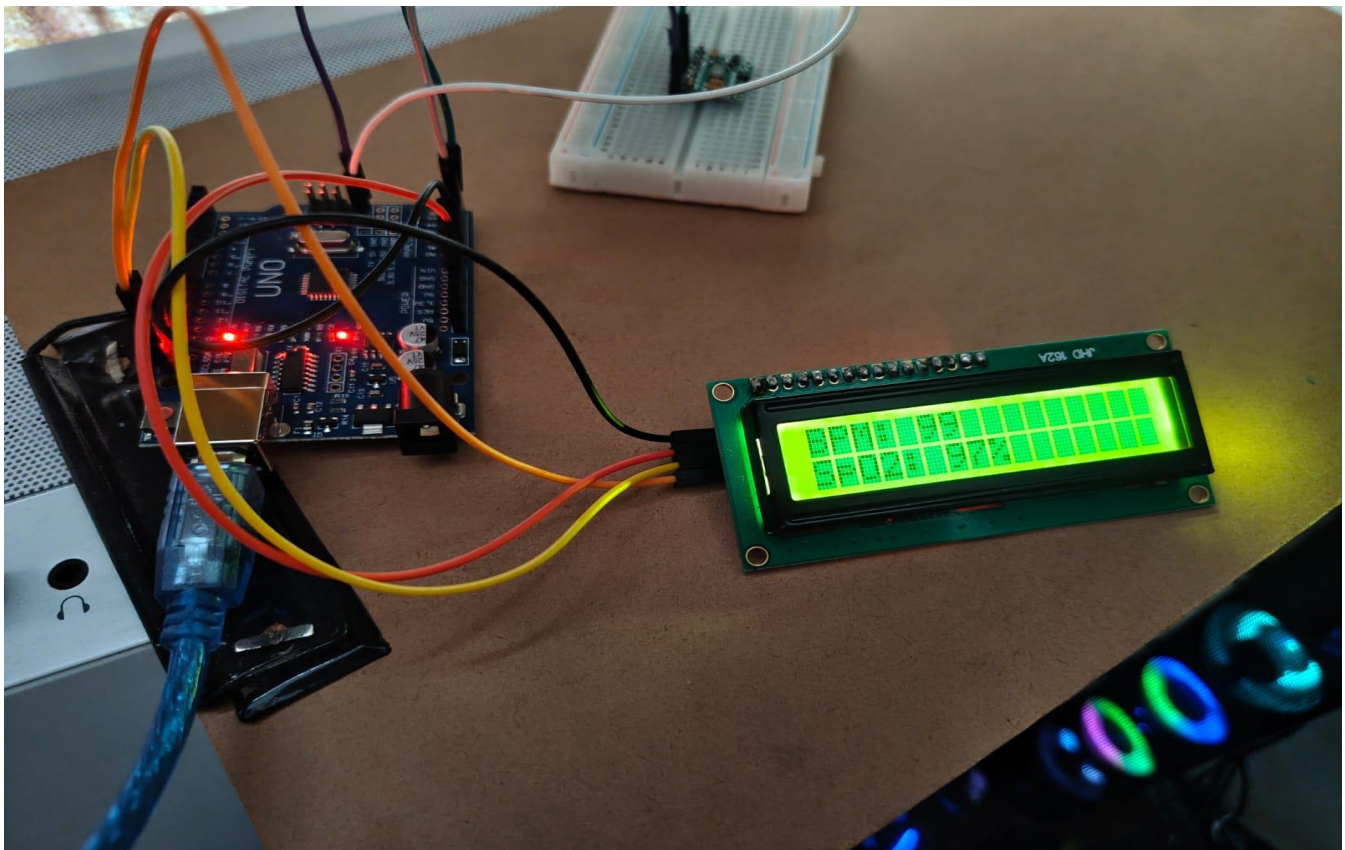
Chapter 1

Introduction

The Patient Health Monitoring System is an IoT-based project that continuously monitors a patient's vital signs, specifically heart rate (BPM) and oxygen saturation (SpO2).

Using the MAX30102 sensor with an Arduino microcontroller, the system displays real-time data on a 16x2 LCD screen, allowing caregivers to monitor patient health easily. This approach is highly effective for cost-effective and continuous care.

This system provides a cost-effective and reliable solution for continuous patient care, helping detect health anomalies promptly and reducing the need for constant manual supervision. Its compact design and real-time alerts make it suitable for hospitals, home healthcare, and elderly care applications.





Chapter 2

Brief Literature Survey

Review of IoT Applications in Health Monitoring 2.1

In recent years, many researchers have explored IoT applications in healthcare, focusing on systems that provide continuous and real-time health monitoring. IoT-based monitoring systems have been designed to measure vital parameters such as heart rate, temperature, and oxygen saturation using sensors connected to microcontrollers. These systems send data wirelessly for real-time observation and emergency alerts, demonstrating high accuracy and low cost in various implementations ,

Hardware Selection and Validation using MAX30102 2.2

Our project follows a similar fundamental concept but simplifies it for educational use by using the MAX30102 sensor with Arduino Uno. The data for BPM and SpO2 are processed and displayed locally on a 16x2 LCD display using I2C communication, allowing the user to see live readings. The integration of the MAX30102 with Arduino has been proven reliable for capturing and processing physiological data accurately, as shown by Lodi et al. [2], Vasam et al. [3] and Lin et al. [4]. The core methodology and sensor integration are inspired by standard electronic design practices and published academic work in this rapidly evolving field.



Chapter 3

Problem Formulation

Need and Significance of Proposed Work

3.1 The Need for Continuous and Affordable Health Monitoring

The main problem this project aims to solve is the need for a **cheap, easy-to-use, and always-on way** to check patient vital signs when they are outside of expensive hospitals. Current manual monitoring methods have major weaknesses: they take a lot of time, people can easily make mistakes when writing down data, and they completely fail to give a continuous flow of data that is necessary for doctors to step in quickly if something goes wrong.

3.2 Significance and Broad Usefulness of the System

The importance of this system is that it directly fixes these problems. It uses simple technology to make high-quality monitoring possible for almost anyone. The key benefits of the proposed work are:

- **Enabling Continuous Monitoring:**
 - The system provides **constant, uninterrupted tracking** of Heart Rate (BPM) and Oxygen Saturation (SpO₂). Instead of checking a patient every few hours, this system checks them every minute. This continuous stream of data helps doctors spot slow changes or sudden, dangerous drops in health much faster.
- **Cost-Effectiveness:**
 - Our solution offers a **low-cost, easy-to-access alternative** to large hospital machines. This makes essential patient care available in places like a patient's home, or for the monitoring of elderly family members, where expensive commercial equipment is not an option.
- **Reducing Staff Burden:**
 - By automating the continuous vital sign checks, the system greatly **minimizes the time** that nurses or caregivers must spend on manual checks. This frees up staff to focus their time and effort on more critical patient needs and direct care, improving the overall quality of support.
- **Workout Help and Broad Utility:**
 - Beyond medical use, the device has a practical, everyday application. It can be used by **fitness enthusiasts to track their heart and oxygen levels during workouts**. This allows users to monitor their performance, ensure they are training safely, and optimize their exercise routine using real-time physiological data.



Chapter 4 Objectives

Objectives of the Proposed Work

The primary goal of this project is to develop an **IoT-Based Patient Health Monitoring System** that is both affordable and capable of providing reliable, real-time physiological data. The specific, measurable objectives are as follows:

4.1 To build a working IoT-Based Patient Health Monitoring System.

This means we plan to create a full, functional machine that successfully connects all the hardware parts and the code to measure and show health data. Our main aim is to build a complete product, not just test the individual pieces.

4.2 To accurately measure and process Heart Rate (BPM) and Oxygen Saturation (SpO2) using the MAX30102 sensor.

Our project must collect health data correctly. We need to write code for the Arduino that can take the raw light signals from the MAX30102 sensor and turn them into accurate numbers for heart rate and blood oxygen level.

4.3 To connect the MAX30102 sensor and display screen to an Arduino microcontroller..

This is the technical goal of making sure all the parts can talk to each other. We must use the **I2C method** to reliably link the sensor (input), the Arduino (brain), and the LCD screen (output), so the system runs smoothly without connection errors.

4.4 To display the monitored vital signs in real-time on a 16x2 LCD screen.

The system must immediately update the measured BPM and SpO2 numbers on the screen. This allows the person watching the patient to see the data right away, locally, without needing an internet connection or another computer.

4.5 To prove the system is a cheap and trustworthy way to monitor patients continuously.

After building the prototype, we need to test it thoroughly. We will check the system's measurements against real medical equipment to confirm that our low-cost project works reliably and can be trusted for continuous health monitoring in places like a patient's home.

Chapter 5

Methodology/Planning of Work

Project Methodology

The project methodology is divided into the following key phases, ensuring a systematic development process:

Phase 5.1: Hardware Selection and Interfacing

This phase involves the acquisition and physical connection of all components:

Acquire the Arduino Uno microcontroller and MAX30102 sensor.

Connect the MAX30102 sensor and the 16x2 LCD display to the Arduino, utilizing I2C communication for the display unit.

Phase 5.2: Software Development (Coding)

The software development is focused on data acquisition and processing:

Install necessary Arduino libraries including

<Wire.h>, <LiquidCrystal_I2C.h>, and "MAX30105.h".

Write the primary code to initialize and read raw data from the sensor.

Implement digital signal processing techniques and algorithms to accurately calculate BPM and SpO2 values from the raw sensor data.

Phase 5.3: Real-Time Data Display and Testing

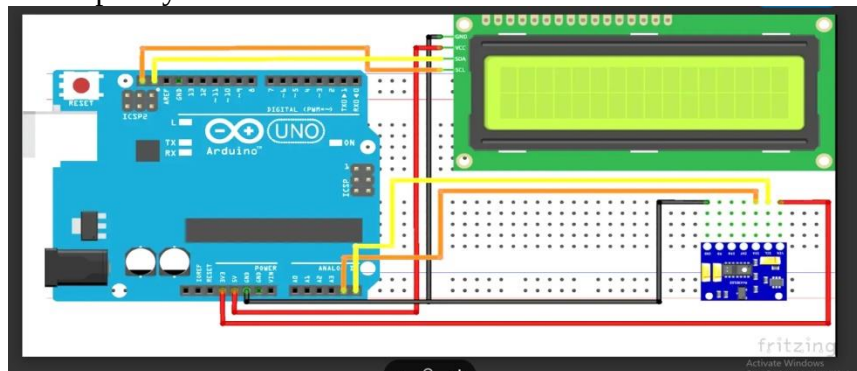
The final stage involves output validation and system verification:

Program the Arduino to send the calculated vital signs to the

LCD screen for real-time output.

Test the complete system under various conditions by placing a finger on the MAX30102 sensor to monitor real-time readings.

Calibrate the system's readings against a commercial medical-grade pulse oximeter to ensure the accuracy and reliability of the developed system.



MAX30102 Sensor to Arduino Uno:

VCC → 3.3V or 5V on Arduino

GND → GND on Arduino

SCL → A5 (or SCL pin) on Arduino

SDA → A4 (or SDA pin) on Arduino

16x2 LCD with I2C Module to Arduino Uno:

VCC → 5V on Arduino

GND → GND on Arduino

SDA → A4 (or SDA pin) on Arduino

SCL → A5 (or SCL pin) on Arduino



Facilities required for proposed work

Resources Required

The successful completion of this project requires the following facilities and resources:

Hardware Components

- **MAX30102** Pulse Oximetry and Heart Rate Sensor.
- **Arduino Uno** Microcontroller Board.
- **16x2 LCD Display** with I2C module.
- **Power supply** (USB cable or HYW Battery).
- **Jumper wires** and **Breadboard** for connections and prototyping.
- **Laptop/PC** with Arduino IDE for coding

Software and Development Environment

- **Arduino IDE** (Integrated Development Environment).
- **C/C++** programming environment.
- Necessary sensor communication and I2C display control libraries, specifically `<Wire.h>`, `<LiquidCrystal_I2C.h>`, and `"MAX30105.h"`.





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